

Minecarts cause chunks to be corrupted/reset when loading worlds from before 1.5 (13w02a)

MC-232104 Datafixer

Status	Resolved / Fixed
Affected	20w21a, 1.17.1, 21w39a, 1.18.1, 1.18.2, 1.19.1 Release Candidate 2 (+2 more)
Fixed in	1.20.5 Pre-Release 4
Reported	2021-07-14

STEPS TO REPRODUCE

1. Launch Minecraft Java Edition 1.4.7.
2. Create a new world.
3. Place a minecart in the world.
4. Save the world and quit to the title screen.
5. Launch Minecraft Java Edition 1.18.2 and load the world.
6. Observe that the chunk containing the minecart fails to load and is replaced with newly generated terrain, resetting the chunk.

OBSERVED BEHAVIOUR

Actual behavior

When a world saved in Java Edition **13w01b or earlier** contains a minecart, loading that world directly in **1.18.2** fails for each affected chunk.

The chunk data fixer encounters the legacy minecart entity ID and throws an exception while converting the chunk's entity data. For example:

```
```text
```

```
Couldn't load chunk [-19, 14]
```

```
java.lang.ClassCastException:
```

```
class com.mojang.serialization.DataResult cannot be cast to class
```

```
com.mojang.datafixers.util.Pair
```

```
```
```

After the load failure, the affected chunk is replaced with a newly generated chunk. The original terrain, blocks, structures, entities, and other chunk data are not preserved; the minecart is not merely removed. In the example world, chunk `[-19, 14]` and other chunks containing legacy minecarts appear reset when opened in 1.18.2, while chunks without such minecarts remain unaffected.

EXPECTED BEHAVIOUR

When a world created in Java Edition 1.4.7 or another version before 13w02a is loaded in a current version, all chunks should load successfully regardless of whether they contain minecarts. The affected chunks should retain their terrain, blocks, structures, and other data, with the minecarts preserved or safely converted as appropriate, and upgrading the world should not replace any chunks with newly generated terrain.

ENVIRONMENT

Minecraft Java Edition Versions: 20w21a, 1.17.1, 21w39a, 1.18.1, 1.18.2, 1.19.1 Release Candidate 2, 1.20.2, 24w12a (includes snapshots)

ORIGINAL DESCRIPTION

May relate to

MC-134115

Resolved

due to 1.13 changes being the current suspect.

The bug

In 13w02a, changes were made to the minecart entity ID. World saves from this version onwards can be safely upgraded to the current version (1.18.2). However, worlds from 13w01b or earlier (back to 12w07a) will cause any chunk that contains a minecart to be completely reset.

How to reproduce

This is as simple as creating a world in 13w01b or earlier, placing a minecart, and then loading that world in the latest version.

Example case

The following is a link to a world save last played in Java Edition 1.4.7 (the file size is rather large so it may be worth making a dedicated world yourself):

http://www.mediafire.com/file/pp1h388kzu2n7p7/Dataless822s_Survival_World_v1.0.zip/file

When loaded up and played in version 1.4.7, it plays completely fine. However, when loaded in the latest supported version, several chunks in the world fail to load, and are replaced with newly generated chunks. In pre-13w02a, the game log also produces errors indicating this:

Example output

Couldn't load chunk [-19, 14]

```
java.lang.ClassCastException: class com.mojang.serialization.DataResult cannot be cast to class
com.mojang.datafixers.util.Pair (com.mojang.serialization.DataResult and com.mojang.datafixers.util.Pair are in
unnamed module of loader 'app')
```

```
at com.mojang.datafixers.optics.Proj2.view(Proj2.java:7)
```

```
at com.mojang.datafixers.optics.Lens.lambda$null$0(Lens.java:48)
```

```
at com.mojang.datafixers.optics.Affine$Instance.lambda$null$0(Affine.java:45)
```

```
at com.mojang.datafixers.optics.Optics$5.preview(Optics.java:117)
```

```
at
```

```
com.mojang.datafixers.types.templates.TaggedChoice$TaggedChoiceType$2.capPreview(TaggedChoice.java:283)
```

```
at com.mojang.datafixers.types.templates.TaggedChoice$TaggedChoiceType$2.preview(TaggedChoice.java:278)
```

```
at com.mojang.datafixers.types.templates.TaggedChoice$TaggedChoiceType$2.preview(TaggedChoice.java:271)
```

```
at com.mojang.datafixers.optics.Affine.lambda$null$2(Affine.java:34)
```

```
at com.mojang.datafixers.optics.Traversal$Instance$1.lambda$wander$0(Traversal.java:34)
```

```
at com.mojang.datafixers.util.Pair$Instance.traverse(Pair.java:90)
```

```
at com.mojang.d
```

MANUAL RESULT

| | |
|------------------|--------------------|
| Version used | |
| Reproduced? | YES / NO / PARTIAL |
| Crash file | |
| Notes | |
| | |
| Tested by / date | |

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-232104>